

PRODUCT REQUIREMENTS DOCUMENT

AI Customer Service & Quotation Desk

Document Type: Product Requirements Document

Product Stage: MVP

Product Category: AI Powered Logistics Customer Operations

Primary Market: Freight Forwarders and Logistics Companies

Initial Users: Customer Service Representatives

Initial Channels: Email and WhatsApp

AI Model: Controlled AI Assistance with Human Approval

1. Product Overview

1.1 Product Name

AI Customer Service & Quotation Desk

Working name for the MVP.

The product is a lightweight AI workspace designed to help logistics customer service teams manage customer enquiries, organize active customer situations, process RFQs, compare quotations, prepare customer quotes, manage follow ups, and identify exceptions requiring attention.

The product is intentionally not designed to replace a TMS or become a complete logistics management platform.

Its purpose is to improve the daily customer service workflow around enquiries and quotations.

2. Product Vision

Create a simple AI powered workspace that helps logistics customer service representatives answer five critical questions every day:

What customer situations need my attention?

What is happening with each customer?

What information is missing?

What do the available agent quotations actually mean?

What should I do next?

The product connects customer communication with RFQ and quotation activity so that customer service representatives do not have to manually reconstruct the workflow from emails, WhatsApp messages, PDFs, spreadsheets and separate notes.

3. Problem Statement

Logistics customer service representatives frequently manage customer enquiries, RFQs, agent communications, quotations and follow ups across several channels.

Information may arrive through:

- Email
- WhatsApp
- PDF quotations
- Excel files
- Word documents
- Images

- Spreadsheets
- Internal messages
- Existing logistics software

A single customer request can therefore become a fragmented process.

The representative may need to:

- Understand what the customer wants
- Identify missing shipment information
- Request clarification
- Request rates from agents or suppliers
- Receive quotations in different formats
- Manually compare charges
- Identify missing charges
- Calculate markup
- Prepare a customer quotation
- Obtain approval
- Send the quotation
- Follow up
- Track the customer's response
- Manage unexpected issues

The core problem is not simply the number of tasks.

The problem is that the information required to complete these tasks is fragmented across conversations, documents and workflows.

Core Problem Statement

Logistics customer service representatives need a simple way to understand customer requests, organize outstanding work, compare received quotation information consistently, identify important differences and determine the appropriate next action without manually piecing information together across multiple channels and documents.

4. Product Opportunity

The MVP combines two closely connected areas:

Customer Service Intelligence

Helps the representative understand:

- What is happening
- What has already happened
- What information is missing
- What is waiting
- What requires attention
- What should happen next

RFQ and Quotation Intelligence

Helps the representative understand:

- What each agent quoted
- What charges are included
- What charges are excluded
- What information is missing

- How quotations compare
- Which quotation requires further clarification
- Which option appears commercially suitable based on available information

The core workflow becomes:

Customer Enquiry → Requirements → RFQ → Agent Responses → Comparison → Recommendation → Markup → Customer Quote → Follow Up → Outcome

5. Product Goals

Primary Goals

Goal 1: Reduce information fragmentation

Bring customer communication and quotation activity into a connected workspace.

Goal 2: Reduce manual RFQ comparison

Extract and normalize quotation information received in different formats.

Goal 3: Improve customer follow up

Ensure important quotations and customer situations do not remain unattended.

Goal 4: Improve quotation visibility

Allow representatives to quickly understand the status and quality of each quotation.

Goal 5: Support better commercial decisions

Provide structured comparison and recommendations while keeping commercial decisions under human control.

Goal 6: Reduce repetitive administrative work

Allow AI to handle information extraction, summarization, classification and drafting.

6. Non Goals

The MVP will not attempt to become:

- A complete TMS
- A freight marketplace
- A carrier booking platform
- A customs management platform
- An accounting platform
- An invoicing platform
- A full CRM
- A warehouse management system
- An advanced transportation planning system
- An autonomous procurement system
- An autonomous negotiation system
- An autonomous pricing system
- A market rate prediction platform
- A complete shipment tracking platform

7. Target Users

Primary User

Logistics Customer Service Representative

Responsible for:

- Receiving enquiries
- Communicating with customers
- Requesting information
- Requesting rates
- Comparing quotations
- Preparing customer quotations
- Following up with customers
- Coordinating internally

Secondary Users

Sales or Business Development

Uses the system to understand:

- Customer opportunities
- Quote status
- Follow ups
- Customer responses
- Lost opportunities

Operations Staff

Uses the system to understand:

- Customer requirements
- Accepted quotations
- Outstanding information
- Exceptions

Customer Service Manager

Uses the system to understand:

- Team workload
- Outstanding enquiries
- Quote status
- Follow up performance
- Exceptions
- Operational bottlenecks

Company Owner or General Manager

Uses summarized information for:

- Customer activity
- Quote outcomes
- Operational visibility
- Commercial performance

8. Primary User Journey

The core customer journey is:

1. Customer sends enquiry

↓

2. System captures the enquiry

↓

3. AI understands the request

↓
4. System identifies missing information
↓
5. Representative obtains required information
↓
6. RFQ is created
↓
7. Agent quotations are received
↓
8. AI extracts quotation information
↓
9. Quotations are normalized
↓
10. System compares quotations
↓
11. System identifies missing or unusual information
↓
12. AI provides recommendation
↓
13. Company markup rules are applied
↓
14. Customer quotation is prepared
↓
15. Human approves quotation
↓
16. Quotation is sent
↓
17. Follow up is created
↓
18. Customer responds
↓
19. Situation becomes accepted, negotiated, rejected or expired
↓
20. Outcome is recorded

9. Product Architecture

The MVP consists of six major modules:

Unified Customer Enquiry Desk

Customer Situation Intelligence

RFQ Intelligence

Quotation Comparison Intelligence

Markup and Customer Quotation

Follow Up and Exception Management

10. Module 1: Unified Customer Enquiry Desk

Purpose

Capture and organize customer enquiries received through supported communication channels.

Initial Channels

- Email
- WhatsApp

Functional Requirements

The system shall:

Capture incoming customer messages.

Identify the customer.

Identify the enquiry.

Group related messages.

Detect shipment requirements.

Extract available shipment information.

Identify missing information.

Summarize the customer request.

Create or update a Customer Situation.

Recommend the next action.

Information Extraction

Where available, the system should identify:

- Customer name
- Company
- Contact information
- Origin
- Destination
- Cargo type
- Quantity
- Weight
- Dimensions
- Equipment type
- Transport mode
- Incoterm
- Required service
- Shipment timeline
- Delivery requirement
- Special instructions
- Other relevant information

Example

Customer message:

> Please give me a quote for 5 cartons from Guangzhou to Lagos.

The system may identify:

Origin: Guangzhou

Destination: Lagos

Quantity: 5 cartons

Potentially missing:

- Weight
- Dimensions
- Cargo description
- Required mode
- Delivery requirement

The system should not invent missing information.

11. Module 2: Customer Situation Intelligence

Purpose

Create a structured view of what is happening with each active customer request.

The fundamental working object is the:

Customer Situation

A Customer Situation represents a specific customer matter requiring action.

Examples:

- Import shipment quotation
- Export shipment quotation
- Rate request
- Customer clarification
- Quote negotiation
- Delivery issue
- Unexpected charge
- Agent issue

Customer Situation Fields

Each situation should contain:

- Customer
- Contact
- Situation type
- Shipment details
- Related conversations
- Related RFQs
- Agent quotations
- Customer quotation
- Current status
- Missing information
- Outstanding actions
- Assigned representative
- Due date
- Recommended next action
- Exceptions
- Final outcome

Situation Status

Possible statuses include:

New

Information Required
RFQ In Progress
Quotation Received
Quotation Under Review
Customer Quote Pending Approval
Quote Sent
Follow Up Due
Negotiation
Accepted
Rejected
Expired
Closed

12. Module 3: RFQ Intelligence

Purpose

Help representatives collect, understand and structure quotations received from agents, carriers, suppliers or other service providers.

Supported Inputs

The MVP should support quotation information received through:

- Email
- WhatsApp
- PDF
- Excel
- Word
- Images where technically supported

RFQ Extraction

The system should extract:

Shipment Information

- Origin
- Destination
- Transport mode
- Equipment
- Cargo information
- Weight
- Volume
- Quantity

Commercial Information

- Freight charge
- Origin charges
- Destination charges
- Handling charges
- Documentation charges
- Delivery charges
- Customs related charges where applicable

- Other charges
- Currency
- Payment terms
- Rate validity
- Transit time

Scope Information

- Included
- Excluded
- Unknown

13. RFQ Normalization

Different agents may provide quotations in different formats.

The system should normalize information into a consistent structure.

Example:

Parameter	Agent A	Agent B	Agent C
Ocean Freight	\$1,850	\$1,720	\$1,900
Origin Charges	\$420	\$600	\$390
Destination Charges	\$650	Unknown	\$580
Transit	35 days	32 days	38 days
Validity	7 days	5 days	10 days

The system must distinguish:

Included

Excluded

Unknown

Unknown must never automatically be treated as zero.

14. Module 4: Quotation Comparison Intelligence

Purpose

Help representatives compare quotations based on the complete commercial and service picture rather than headline freight cost alone.

Comparison Dimensions

Cost

- Freight
- Origin charges
- Destination charges
- Handling
- Documentation
- Delivery
- Other applicable charges
- Estimated total

Service

- Transit time
- Route
- Service type

- Scope
- Commercial
- Currency
- Validity
- Payment terms
- Minimum charges
- Conditions
- Completeness
- Missing charges
- Unclear charges
- Excluded charges
- Conflicting information

15. Missing Charge Detection

The system should identify important missing information.

Example:

Agent A:

Total identifiable cost: \$2,920

Agent B:

Headline freight: \$2,700

Destination charges: Unknown

The system should not conclude that Agent B is cheaper.

Instead:

Agent B has a lower stated freight cost, but destination charges are unavailable. Total cost cannot currently be compared with confidence.

16. Quotation Confidence

Every comparison should include a confidence level.

High Confidence

Major parameters are available and comparable.

Medium Confidence

Some interpretation or clarification is required.

Low Confidence

Important information is missing or inconsistent.

Example:

Low comparison confidence

Destination charges are missing from two quotations.

17. Quotation Recommendation Engine

The system may provide a recommendation based on available information.

The recommendation should consider:

- Estimated total cost
- Completeness
- Transit time

- Service scope
- Validity
- Commercial conditions
- Historical agent performance where sufficient data exists

The recommendation must explain its reasoning.

Example:

Recommendation

Agent A appears commercially suitable based on the current information because the quotation contains a complete cost structure, confirmed destination charges and competitive total cost.

Consideration

Agent B has a lower stated freight rate, but destination charges are unavailable. Further clarification is required before treating it as the lower total cost option.

The system must not simply label an agent as:

Best Agent

without explaining the basis.

18. Historical Agent Intelligence

Historical information may be used when sufficient reliable data exists.

Possible indicators:

- Previous response reliability
- Quote accuracy
- Previous service issues
- Transit performance
- Customer complaints
- Rate competitiveness
- Frequency of quotation revisions

Historical intelligence is not required for the earliest MVP.

The MVP should primarily use information contained in the current RFQ.

19. Module 5: Markup Recommendation

Purpose

Apply company defined pricing rules to the selected agent cost.

The company should define rules such as:

- Percentage markup
- Fixed markup
- Minimum margin
- Combined pricing rules

Example

Agent cost:

\$2,920

Markup:

12%

Recommended customer price:

\$3,270.40

The representative can:

- Accept
- Edit
- Override

The AI does not independently define the company's pricing policy.

20. Customer Quotation Workspace

Once the representative selects a quotation and approves the pricing:

The system prepares the customer quotation.

The quotation should contain:

- Customer information
- Shipment information
- Service
- Route
- Charges
- Markup
- Final customer price
- Currency
- Validity
- Transit time
- Terms
- Relevant conditions

The quotation remains subject to human approval before being sent.

21. Module 6: Follow Up and Exception Management

Follow Up

The system should track:

Quote Sent

↓

Awaiting Customer

↓

Follow Up Due

↓

Customer Responded

↓

Negotiation

↓

Accepted / Rejected / Expired

Follow Up Intelligence

The system should identify:

- Quotes requiring follow up
- Overdue follow ups
- Customer responses requiring attention
- Quotes approaching expiry

- Customers who have not responded
- Active negotiations

22. Lightweight Exception Management

The MVP should provide simple exception tracking rather than attempting to build a full incident management platform.

Examples:

- Customer complaint
- Agent delay
- Missing information
- Unexpected charge
- Service mismatch
- Internal approval delay
- Conflicting quotation information
- Customer requirement change

Each exception should record:

- What happened
- Customer affected
- Situation affected
- Responsible person
- Next action
- Due date
- Status

23. Daily Customer Service Dashboard

The primary dashboard should answer:

What needs my attention today?

Example:

Today

3 Urgent

Customer response required.

7 Follow Ups

Customer quotations awaiting response.

4 Pending Agent Responses

RFQs still awaiting quotations.

2 Quotes Awaiting Approval

Customer quotations prepared.

2 Exceptions

Potential service or quotation issues.

The dashboard should prioritize actionable work rather than simply displaying information.

24. AI Assistant

The AI layer should support:

Understanding

Extract information from messages and documents.

Summarization

Summarize customer situations and quotation information.

Detection

Identify missing information, inconsistencies and exceptions.

Recommendation

Suggest appropriate next actions.

Drafting

Draft customer communications and internal messages.

Comparison

Normalize and compare quotations.

Explanation

Explain why an option has been recommended.

25. AI Autonomy Framework

The product will use controlled autonomy.

Level 1: Understand

AI reads and structures information.

Level 2: Recommend

AI recommends actions.

Level 3: Draft

AI prepares communications and quotations.

Level 4: Human Approval

Human confirms consequential actions.

The MVP will not allow AI to independently:

- Commit pricing
- Select suppliers without oversight
- Negotiate commercially
- Send binding quotations
- Book shipments
- Make contractual commitments

26. Human Approval Requirements

Human approval is required before:

- Customer quotation is sent
- Pricing is committed
- Supplier or agent selection becomes commercially binding
- Negotiation position is communicated
- Any action creates significant financial or contractual consequences

27. Core Data Model

The MVP should conceptually use the following relationship:

Customer

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Customer Situation



Conversation



RFQ



Agent



Agent Quotation



Quotation Comparison



Markup Recommendation



Customer Quotation



Follow Up



Outcome

Supporting objects:

- User
- Action
- Exception
- AI Recommendation
- Attachment

28. Core Workflow States

Customer Situation

New → Information Required → RFQ In Progress → Quotation Received → Review →
Customer Quote → Quote Sent → Follow Up → Negotiation → Outcome

RFQ

Draft → Sent → Awaiting Response → Partial Responses → Responses Received →
Comparison → Clarification Required → Complete

Agent Quotation

Received → Extracted → Validated → Comparable → Needs Clarification → Selected / Not
Selected

Customer Quotation

Draft → Approval Required → Approved → Sent → Follow Up → Accepted / Rejected / Expired

29. MVP Functional Requirements

Customer Management

The MVP must support:

- Customer profiles
- Contact information

- Customer situations
- Conversation grouping
- Customer history
- Active quotation visibility

Enquiry Management

The MVP must support:

- Email intake
- WhatsApp intake
- AI extraction
- Missing information detection
- Conversation summarization
- Customer situation creation

RFQ Management

The MVP must support:

- RFQ creation
- RFQ information extraction
- Agent quotation ingestion
- PDF extraction
- Basic Excel extraction
- Basic document extraction
- Quotation normalization

Comparison

The MVP must support:

- Charge comparison
- Total identifiable cost
- Included charges
- Excluded charges
- Unknown charges
- Missing information detection
- Confidence level
- Recommendation explanation

Commercial

The MVP must support:

- Configurable markup rules
- Markup recommendation
- Customer quotation preparation
- Human approval

Follow Up

The MVP must support:

- Quote status
- Follow up dates
- Outstanding actions
- Customer response status
- Quote expiry

- Outcome tracking

Exceptions

The MVP must support:

- Exception creation
- Exception category
- Owner
- Due date
- Next action
- Status

30. Out of Scope for MVP

The following are explicitly excluded:

- Full TMS functionality
- Automated shipment booking
- Customs management
- Accounting
- Invoicing
- Automated procurement
- Full freight marketplace
- Advanced carrier integrations
- Dynamic market pricing
- Predictive freight rates
- Autonomous negotiation
- Autonomous pricing
- Fully autonomous customer communication
- Advanced predictive analytics
- Advanced shipment tracking
- Complex enterprise reporting

31. User Stories

Customer Service

As a customer service representative, I want to see all active customer situations in one place so that I know what requires attention.

As a customer service representative, I want the system to summarize customer conversations so that I do not have to reread long message threads.

As a customer service representative, I want the system to identify missing shipment information so that I know what to request from the customer.

As a customer service representative, I want the system to show what is waiting on me and what is waiting on someone else.

RFQ

As a customer service representative, I want quotation information extracted from documents and messages so that I do not manually retype every rate.

As a customer service representative, I want different agent quotations converted into a common structure so that I can compare them.

As a customer service representative, I want the system to identify missing charges so that I do not accidentally compare incomplete quotations.

As a customer service representative, I want the system to explain its recommendation so that I can make the final commercial decision.

Customer Quotation

As a customer service representative, I want the system to calculate markup according to company rules so that quotation preparation is faster.

As a customer service representative, I want to review and edit the proposed customer quotation before it is sent.

Follow Up

As a customer service representative, I want to know which quotations require follow up so that opportunities do not remain unattended.

As a manager, I want visibility into overdue follow ups so that I can identify operational gaps.

32. Example End to End Scenario

A customer sends a WhatsApp message:

> We need a quote for one 20 foot container from China to Lagos.

The system creates a Customer Situation.

It identifies:

- Origin: China
- Destination: Lagos
- Equipment: 20 foot container
- Service: Import freight

It identifies missing information.

The representative obtains the required details.

The RFQ is created.

Three agents respond.

Agent A sends a PDF.

Agent B sends an Excel quotation.

Agent C responds through WhatsApp.

The system extracts the information.

It normalizes the quotations.

It identifies:

Agent A:

\$2,920 identifiable total

Agent B:

\$2,700 freight plus unknown destination charges

Agent C:

\$3,050 identifiable total

The system flags Agent B.

Issue: Destination charges are unavailable.

The system compares the quotations based on identifiable cost, completeness, service and validity.

It recommends Agent A based on the available information.

The representative reviews the recommendation.

A 12% markup is applied according to company rules.

The customer quotation is prepared.

The representative approves it.

The quotation is sent.

The system creates a follow up.

If the customer does not respond, the situation appears in the follow up queue.

If the customer negotiates, the situation changes to Negotiation.

If accepted, the situation becomes Accepted.

33. Product Success Metrics

Customer Service Efficiency

- Average time to understand a customer situation
- Average time from enquiry to RFQ creation
- Outstanding actions completed
- Follow ups completed on time
- Number of unresolved situations

RFQ Efficiency

- Average time to process an RFQ
- Average time to compare quotations
- Extraction accuracy
- Number of manual corrections
- Missing information detection accuracy

Commercial Efficiency

- Average time from RFQ receipt to customer quotation
- Markup recommendation acceptance rate
- Markup override rate
- Quote approval time
- Quote conversion rate

Product Usage

- Daily active representatives
- Customer situations created
- RFQs processed
- Quotations compared
- Customer quotations prepared
- Follow ups completed
- AI recommendations accepted or overridden

34. AI Quality Metrics

The AI should be evaluated on:

Extraction Accuracy

How accurately does it extract shipment and quotation information?

Classification Accuracy

How accurately does it classify customer situations?

Missing Information Accuracy

How accurately does it identify information that is genuinely missing?

Comparison Accuracy

How accurately does it normalize and compare charges?

Recommendation Quality

How often do representatives agree with the recommendation?

Human Correction Rate

How often does the representative need to correct AI generated information?

Explainability

Can the representative understand why the system produced the recommendation?

35. Critical Product Risks

Risk 1: Existing Software Already Solves the Problem

Freight forwarding software and newer AI products already provide capabilities around quoting, RFQs, customer management and workflow automation.

The MVP therefore must prove that it creates meaningful value for its target users.

Risk 2: Integration Complexity

Email and WhatsApp workflows can be fragmented.

The product must avoid requiring complex implementation at the beginning.

Risk 3: Poor Document Quality

Agent quotations may contain inconsistent formats, images, tables and unclear charges.

The product must clearly communicate uncertainty rather than inventing values.

Risk 4: Commercial Liability

Incorrect pricing or quotation recommendations can create financial consequences.

Human approval therefore remains mandatory.

Risk 5: Workflow Adoption

Representatives may continue using WhatsApp, email and spreadsheets even after adopting the product.

The product must fit into the existing workflow rather than requiring an immediate complete workflow replacement.

Risk 6: Overbuilding

Adding full TMS, CRM, tracking, procurement and accounting capabilities would significantly increase complexity.

The MVP must remain focused on customer service and RFQ intelligence.

36. Security and Data Considerations

The system may process:

- Customer contact information
- Commercial rates
- Agent information

- Shipment information
- Pricing information
- Business correspondence
- Documents

Therefore the product should include appropriate:

- Authentication
- Role based access
- Data isolation between companies
- Secure document storage
- Audit logs
- Access controls
- Data retention controls

AI generated information should remain distinguishable from verified human entered information.

37. Auditability

Important AI actions should be traceable.

The system should record:

- Source document or message
- Extracted information
- AI interpretation
- User correction
- Recommendation
- Human decision
- Final action

Example:

AI extracted destination charge: \$650

Source: Agent quotation PDF.

Representative corrected value: \$680

The system should preserve this change.

38. MVP Acceptance Criteria

The MVP should be considered functional when a representative can complete the following workflow:

Step 1

Receive a customer enquiry.

Step 2

View an AI generated summary.

Step 3

See extracted shipment information.

Step 4

See missing information.

Step 5

Create or update a customer situation.

Step 6

Create an RFQ.

Step 7

Receive multiple agent quotations.

Step 8

Extract quotation information.

Step 9

Normalize quotation information.

Step 10

Compare quotations.

Step 11

Identify missing or unknown charges.

Step 12

Review AI recommendation.

Step 13

Apply company markup rules.

Step 14

Prepare customer quotation.

Step 15

Approve quotation.

Step 16

Track quotation status.

Step 17

Receive customer response.

Step 18

Create and manage follow up.

Step 19

Track the final outcome.

39. MVP Definition of Done

The MVP is ready for controlled pilot testing when:

- Customer enquiries can be captured and structured.
- Customer situations can be created and tracked.
- Conversations can be connected to situations.
- Shipment requirements can be extracted.
- Missing information can be identified.
- RFQs can be created.
- Agent quotations can be processed.
- Common quotation fields can be normalized.
- Charges can be compared.
- Unknown charges are clearly identified.
- AI can provide an explainable recommendation.
- Company markup rules can be configured.
- Customer quotations can be prepared.

- Human approval is required before sending.
- Follow ups can be created and tracked.
- Basic exceptions can be recorded.
- User actions are auditable.
- Representatives can complete the core workflow without relying on the product team for manual intervention.

40. Product Principle

The product should follow one central principle:

Do not automate individual tasks in isolation. Connect the workflow around the customer's intent.

The product is not simply an AI email reader.

It is not simply an RFQ parser.

It is not simply a quotation comparison tool.

It connects:

Customer → Situation → RFQ → Agent Quotations → Comparison → Commercial Decision →

Customer Quote → Follow Up → Outcome

That connected workflow is the foundation of the MVP.

41. Final MVP Definition

AI Customer Service & Quotation Desk

A lightweight AI workspace for logistics customer service teams that connects customer enquiries, RFQs, quotation comparison, customer quotations and follow ups in one daily workflow.

The MVP helps representatives:

Understand the customer.

Know what is missing.

Understand what needs attention.

Process incoming RFQs.

Compare quotations consistently.

Identify incomplete or unclear charges.

Apply company pricing rules.

Prepare customer quotations.

Track follow ups.

Manage basic exceptions.

Keep humans in control of consequential decisions.

The MVP deliberately stops before becoming a full TMS, freight marketplace or autonomous logistics operating system.